5.3 Factorising $z^{2n}+1$

WORKSHEET

The question

STEP III 2013, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q4

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Where are the solutions of $z^{2n} = -1$ in the complex plane? Draw them for $n = 2$ and $n = 4$.

STAGE 1 Conjugate pairs give real quadratics

Covers: the factorisation

Ask yourself Multiply out $(z - e^{i\theta})(z - e^{-i\theta})$ and simplify the middle coefficient.

STAGE 2 The factorisation

Covers: the factorisation

Pause and try List the $2n$ solutions of $z^{2n}=-1$ as conjugate pairs $e^{\pm \mathrm{i}\theta_k}$. What are the θ_k? Why are they all the roots?

STAGE 3 Substituting $z=\mathrm{i}$

Covers: Part (i)

Ask yourself What is $\mathrm{i}^2-2\mathrm{i}\cos\theta+1$? And what is $\mathrm{i}^{2n}+1$ when n is even?

Pause and try Simplify $(-2\mathrm{i})^n$ when n is even.

STAGE 4 When n is odd

Covers: Part (ii)

Ask yourself For odd n, one of the angles θ_k equals $\frac{\pi}{2}$. Which one? Why does that make the substitution $z = \mathrm{i}$ useless as it stands?

Pause and try Divide both sides of the factorisation by $1 + z^2$, then put $z = \mathrm{i}$. Pair up the remaining cosines: how is $\cos\theta_{n+1-k}$ related to $\cos\theta_k$?

What's really going on

The fundamental theorem of algebra

That $z^{2n} + 1$ can be factorised completely into linear factors over $\mathbb{C}$ is a special case of the *fundamental theorem of algebra*: every non-constant polynomial with complex coefficients has a complex root, and so factorises completely into linear factors. Gauss gave the first widely accepted proof in his doctoral thesis of 1799, and returned to the theorem several times with new proofs. A consequence is that every real polynomial factorises into real linear and quadratic factors, because non-real roots come in conjugate pairs, exactly as in Stage 1. The theorem is "algebraic" in name but analytic in substance: every proof uses some property of the real numbers, such as continuity, and you will see one of the slickest proofs in a first course on complex analysis.

Products of sines and cosines

The method gives a whole family of identities. Using z^n-1 in place of $z^{2n}+1$ leads to

$$\prod_{k=1}^{n-1} \sin\frac{k\pi}{n} = \frac{n}{2^{n-1}},$$

which is proved, in its geometric form, in Section 5.4. Identities like these appear in the analysis of vibrating systems (the frequencies of n masses joined by springs involve $\sin\frac{k\pi}{2(n+1)}$) and in counting problems (the number of spanning trees of some graphs, and the number of domino tilings of a rectangle, are given by products of cosines). Kasteleyn's 1961 formula for domino tilings of an $m\times n$ rectangle is a double product of expressions in $\cos^2\frac{j\pi}{m+1}$ and $\cos^2\frac{k\pi}{n+1}$.

Try this next

1. **A heptagonal product.** Use part (ii) with $n=7$ to show that $\cos\frac{\pi}{14}\cos\frac{3\pi}{14}\cos\frac{5\pi}{14}=\frac{\sqrt{7}}{8}$. Then show that $\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{3\pi}{7}=\frac{1}{8}$. (*Hint:* relate the angles using $\cos\theta=\sin(\frac{\pi}{2}-\theta)$, or use the identity $\sin 2x = 2\sin x\cos x$ repeatedly.)

2. **Another substitution.** Substitute $z=1$ in the factorisation to show that
$$\prod_{k=1}^{n}\sin\frac{(2k-1)\pi}{4n} = 2^{\frac{1}{2}-n}.$$

3. **Factorise z^6+1.** Write z^6+1 as a product of three real quadratics, and check your answer by multiplying out.

4. **Sines.** By factorising $z^{2n}-1$ similarly and dividing by z^2-1, show that
$$\prod_{k=1}^{n-1}\sin\frac{k\pi}{2n} = \frac{\sqrt{n}}{2^{n-1}}.$$

Further reading

- **Tristan Needham, *Visual Complex Analysis*** (Oxford University Press). Chapter 2 on roots of unity and polynomials, with beautiful pictures.
- **Benjamin Fine & Gerhard Rosenberger, *The Fundamental Theorem of Algebra*** (Springer). Several proofs of the theorem, from analysis, algebra and topology, written for undergraduates.
- **Paul Nahin, *An Imaginary Tale*** (Princeton University Press). A history of $\sqrt{-1}$, including roots of unity and their uses.

The full guided solution is in Section 5.3:
https://beyondstep.keithmcnulty.org/ch05/z2n-plus-one.html